30 A Geiger-Müller counter is placed near a fresh radioactive sample with a known half-life. It reads an initial total count rate of 200 counts min^{-1}. After four half-lives have elapsed, the total count rate decreases to one-quarter of its initial value.

What is the background count rate?

A 0 counts min^{-1} **B** 13 counts min^{-1} **C** 40 counts min^{-1} **D** 50 counts min^{-1}